

Robust DOA Estimation in Impulsive Noise via Entropy-Adaptive Processing Algorithm

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Abstract—Direction of Arrival (DOA) estimation is a significant branch of array signal processing. In DOA estimation, traditional algorithms for handling impulsive noise rely heavily on impulsive noise information, limiting their applicability to broader impulsive noise environments. To overcome this limitation, the entropy-adaptive processing algorithm (EAPA) is proposed. This algorithm constructs the covariance matrix by incorporating information entropy and Cauchy kernel, enabling effective DOA estimation under impulsive noise conditions without requiring prior noise information. Simulation results demonstrate that the proposed EAPA algorithm significantly outperforms existing methods, achieving high estimation accuracy and robustness in DOA estimation even under conditions of no prior information.

Index Terms—Impulsive noise, information entropy, Cauchy kernel, adaptive, DOA estimation.

I. INTRODUCTION

DIRECTION of Arrival (DOA) estimation is one of the core problems in array signal processing, holding significant importance not only in research but also in engineering applications [1]. The development of the low-altitude economy has further expanded the application demand for DOA estimation in emerging scenarios such as drone monitoring and air traffic management. This technique infers the direction of a signal source using signals received by an array of sensors. With technological advancements, this field has achieved significant improvements in accuracy, computational efficiency, and interference resistance. In particular, subspace methods based on eigenvalue decomposition, such as the multiple signal classification (MUSIC) [2] and the estimating signal parameter via rotational invariance techniques (ESPRIT) [3] developed in the last century, have greatly enhanced the accuracy of DOA estimation, marking a milestone in the field.

However, subspace-based methods typically rely on the assumption of Gaussian noise and perform excellently under

conditions of high generalized signal-to-noise ratio (GSNR) and large number of snapshots. Yet, their performance is significantly constrained in adverse scenarios such as low GSNR and small number of snapshots. Furthermore, noise in real-world environments often exhibits pulsed characteristics, causing traditional DOA estimation algorithms reliant on second-order statistics [4] to experience a sharp decline in performance or even complete failure. Consequently, addressing the DOA estimation problem under impulsive noise conditions has become a prominent research focus in the field [5], [6]. Traditional DOA estimation algorithms are largely based on the assumption of Gaussian noise. When dealing with impulsive noise in real-world environments, the performance of these traditional algorithms deteriorates significantly.

To address the challenges posed by impulsive noise to traditional DOA estimation methods, researchers have proposed various impulsive noise-resistant strategies. The fractional-order low-order moment (FLOM) method proposed in [7] exhibits desirable performance specifically under the condition of characteristic exponent $\alpha > 1$, while its efficacy degrades remarkably in the absence of this constraint. [8] employs the Phased fractional low-order moments (PFLOM) method, but PFLOM computation relies on prior knowledge of the noise distribution (e.g., characteristic parameters). [9] proposed the spatial sign covariance matrix (SCM) algorithm, but its theoretical performance relies on the assumption that noise follows a spherically symmetric distribution. Meanwhile, infinite-norm normalized MUSIC [10] and zero-memory nonlinear (ZMNL) [11] methods introduce effects on the received raw signal.

Recently, based on the characteristics of correlation entropy, scholars have proposed a series of methods to combat impulsive noise. [12] introduced deviation from the median based correlentropy (DMC), which replaces the traditional mean with the median of the signal to calculate deviation and constructs a pseudo-covariance matrix using an exponential kernel function. Building upon this, [13] extended the correlentropy

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concept by incorporating broader nonlinear kernel functions, though its noise suppression effectiveness was significantly reduced. [14] constructed an infinite-norm Cauchy kernel to suppress impulsive noise in received signals, yet this algorithm also requires prior knowledge about the noise.

To address the aforementioned issues, the contribution of this paper are; 1) A real-time DOA estimation method for which no prior noise information is required and adaptive noise suppression capabilities are possessed; 2) Based on the array-received signals, information entropy and kernel functions are utilized to compute weights for each received signal, thereby performing denoising processing; 3) A covariance matrix is then constructed from the denoised signals to validate the effectiveness of the EAPA algorithm. Simulation results demonstrate that this method achieves both high estimation accuracy and robustness in DOA estimation without requiring prior information.

II. DIRECTION OF ARRIVAL MODEL

The uniform linear array (ULA) is one of the most commonly used array structures for DOA estimation. This array consists of M elements arranged along a straight line at equal intervals, with an element spacing of d ($d = \lambda/2$). Consider a scenario where K far-field narrowband signal sources ($K < M$) arrive at different angles impinge on a uniform linear array composed of M omnidirectional sensors. Establish a received signal model with the first element as the reference point. At time t , the received signal vector of the array can be expressed as:

$$\mathbf{Z}(t) = \sum_{k=1}^K \mathbf{a}(\theta_k) s_k(t) + \mathbf{n}(t) = \mathbf{A}(\theta) \mathbf{s}(t) + \mathbf{n}(t), \quad (1)$$

Here, θ_k denotes the angle at which the k th far-field signal source incident onto the uniform line array. $\mathbf{s}(t) = [s_1(t), s_2(t), \dots, s_K(t)]^T$ represents the signal vector for different incident angles. $\mathbf{A}(\theta) = [\mathbf{a}(\theta_1), \mathbf{a}(\theta_2), \dots, \mathbf{a}(\theta_K)]$ denotes the array directional matrix. The direction vector can be represented as $\mathbf{a}(\theta_k) = e^{-j2\pi(M-1)d\sin\theta_k/\lambda}$. $\mathbf{n}(t) = [\mathbf{n}_1(t), \mathbf{n}_2(t), \dots, \mathbf{n}_M(t)]^T$ is a impulsive noise vector obeying α stable distribution [8]. $\mathbf{n}(t) = [\mathbf{n}_1(t), \mathbf{n}_2(t), \dots, \mathbf{n}_M(t)]^T$ denotes the impulsive noise vector.

III. PROPOSED EAPA ALGORITHM

A. Denoising via ∞ -Norm Cauchy Kernel Regression

The presence of impulsive noise causes severe degradation or even complete failure of DOA estimation algorithms based on subspace decomposition, such as the MUSIC algorithm. To restore algorithm effectiveness, preprocessing of the array received signal is essential to suppress impulsive interference.

The infinity norm cauchy kernel algorithm employs the infinity norm cauchy kernel as its core preprocessing mechanism. Here, $\mathbf{Z}(t)$ represents the received signal after denoising, and T denotes the snapshot. After preprocessing, the reconstructed signal's covariance matrix can be expressed as:

$$\tilde{\mathbf{R}}_{\mathbf{Y}} = \{\mathbf{Z}(t) \mathbf{Z}^H(t)\} / T, \quad (2)$$

Using the obtained $\tilde{\mathbf{R}}_{\mathbf{Y}}$, the subspace algorithm yields the estimated angle $\hat{\theta}_k$, where $(\cdot)^H$ denotes the conjugate transpose operation. Under impulsive noise following an α -distribution, $\tilde{\mathbf{R}}_{\mathbf{Y}}$ is unbounded, and effectiveness depends on accurately obtaining the impulsive noise characteristic exponent α . Traditional algorithms such as PFLOM and FLOM require prior knowledge of α .

B. Entropy-Adaptive Processing Algorithm for DOA Estimation

Along with the evolution of DOA estimation techniques toward real-time and adaptive capabilities, higher demands are placed on algorithmic dependence on prior information and computational complexity. Traditional algorithms such as PFLOM, FLOM, and INCK rely on noise feature parameters as prior knowledge and struggle to adapt to the statistical variations of non-stationary impulsive noise, revealing significant limitations in practical applications.

In recent years, the innovative application of information entropy theory in array signal processing has garnered significant attention within the academic community. By quantifying uncertainty characteristics in the signal space, it provides a novel theoretical framework for DOA estimation. To this end, this paper proposes the EAPA algorithm, which establishes a real-time DOA estimation method that requires no prior noise information and possesses adaptive noise suppression capabilities.

The algorithm first constructs the information entropy under the t snapshot based on the signal received by the array H_t :

$$H_t = - \sum_{i=1}^M p_{it} \log(p_{it}), p_{it} = \frac{|\mathbf{z}_i(t)|}{\sum_{k=1}^M |\mathbf{z}_k(t)|}, \quad (3)$$

where $\mathbf{z}_i(t)$ represents the signal information of the i array element under the t snapshot. Construct the kernel function under the t snapshot based on the array received signal $\mathbf{Z}(t)$:

$$\begin{cases} \omega_1(i, t) = 1 / (1 + |\mathbf{z}_i(t)| \|\mathbf{z}(t)\|_{\infty} / (2\pi))^2 \\ \omega_2(i, t) = 1 / (1 + \|\mathbf{z}(t)\|_{\infty}^2 / (2\pi))^2, \end{cases} \quad (4)$$

where $\mathbf{z}(t)$ represents the signal information received by all array elements at the t snapshot, and $\|\cdot\|_{\infty}$ is defined as the maximum absolute value of the vector elements. This enables the construction of the corresponding weight $\mathbf{W}$ for each received signal:

$$\mathbf{W}(i, t) = a\omega_1(i, t) + (1 - a)\omega_2(i, t), \quad (5)$$

where $a = \tanh(\mathbf{H}_t)$, $\tanh(\cdot)$ represents the hyperbolic tangent function. $\tanh(\mathbf{H}_t)$ autonomously weights processing strategies based on entropy, switching between highly impulsive noise and steady noise without preset parameters. The signal $\hat{\mathbf{Y}}(t)$ after denoising is expressed as:

$$\hat{\mathbf{Y}}(t) = \mathbf{W}(i, t) \mathbf{z}(t), \quad (6)$$

Then, construct the covariance matrix $\tilde{\mathbf{R}}_{\text{EPEA}}$ based on the signal information $\hat{\mathbf{Y}}(t)$ after denoising:

$$\tilde{\mathbf{R}}_{\text{EPEA}} = \hat{\mathbf{Y}} \hat{\mathbf{Y}}^H / T, \quad (7)$$

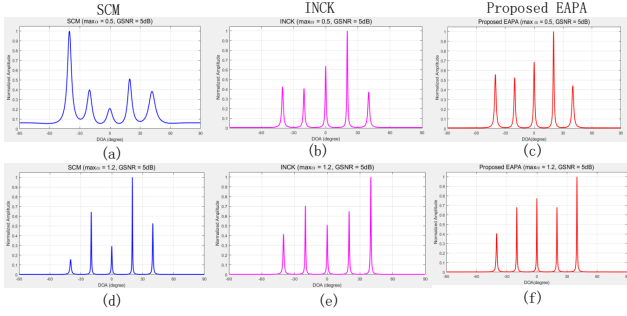


Fig. 1. Performance comparison across multiple data sources.

Finally, applying the subspace method to $\tilde{\mathbf{R}}_{\text{EAPA}}$ yields the corresponding DOA estimates.

IV. SIMULATION EXPERIMENT AND ANALYSIS

When α is stable, the GSNR is commonly used to represent the relative strength of signal and noise energy. This simulation employs the GSNR defined by (8) to characterize the energy ratio between signal and noise, namely:

$$\text{GSNR} = 10 \lg \left(\frac{\mathbb{E} |s(t)|^2}{\gamma} \right), \quad (8)$$

Here, $\mathbb{E} \{s(t)\}^2$ denotes the signal's average probability, while γ represents the noise's dispersion sparsity. This paper employs Monte Carlo simulations to evaluate the performance gap between the proposed algorithm and other methods using curves formed by the root mean square error (RMSE) of the estimated angle $\hat{\theta}_k$. The RMSE is defined as follows:

$$\text{RMSE} = \frac{1}{\text{MC}} \sum_{j=1}^{\text{MC}} \sqrt{\frac{1}{K} \sum_{k=1}^K (\tilde{\theta}_{kj} - \theta_k)^2}, \quad (9)$$

Here, MC denotes the number of Monte Carlo trials, $\tilde{\theta}_{kj}$ represents the precise estimate of the DOA for the k signal source in the j th Monte Carlo process, θ_k indicates the actual angle of the k signal source, and K denotes the total number of signal sources.

We thoroughly validates the effectiveness of the EAPA algorithm through comparison with other algorithms under varying impulsive noise levels, GSNR, and snapshot.

a) Simulation Experiment 1: Comparative experiments of SCM [9], INCK [14], and EAPA under multi-source and different impulsive noise conditions..

In this experiment, five far-field narrowband independent signals with azimuth angles $[-40^\circ, -20^\circ, 0^\circ, 20^\circ, 40^\circ]$, are incident onto a 10-element uniform linear array. The GSNR is set to 5 dB, and $T = 400$. All algorithms employed in this experiment enable effective DOA estimation under impulsive noise conditions. Through MC simulations with MC = 1000 iterations, the DOA estimation accuracy and robustness of the SCM, INCK, and EAPA algorithms were evaluated for impulsive noise characteristic indices $\alpha = 0.5$ and $\alpha = 1.2$.

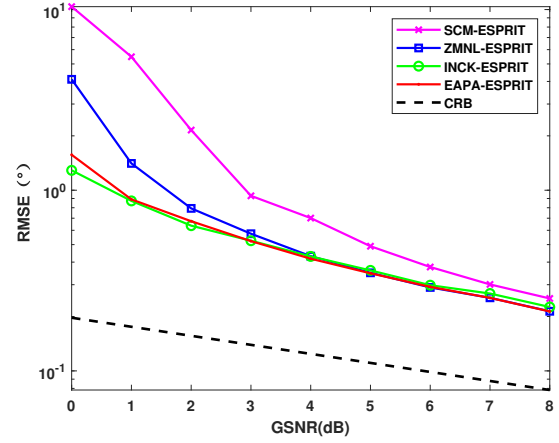


Fig. 2. Performance comparison of four algorithms at different signal-to-noise ratios when $\alpha=0.8$.

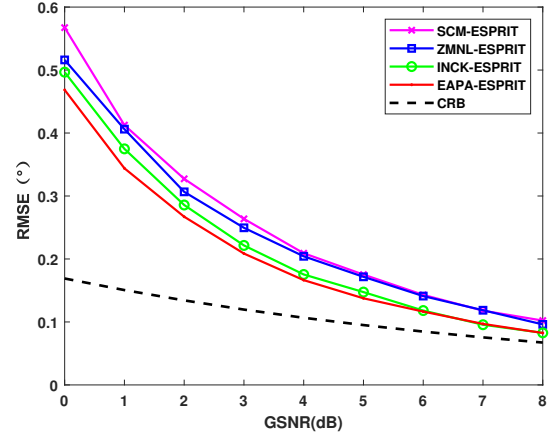


Fig. 3. Performance comparison of four algorithms at different signal-to-noise ratios when $\alpha=1.5$.

As evidenced in Fig. 1, the spatial spectrum peak of the EAPA algorithm exhibits a sharper peak morphology compared to SCM and INCK, intuitively validating the effectiveness and superiority of the EAPA algorithm in DOA estimation. Furthermore, as shown in Fig. 1(c), the algorithm still demonstrates excellent performance under high-pulse conditions with $\alpha = 1.2$. This experiment thus validates the effectiveness of the proposed EAPA algorithm for multi-target DOA estimation in the presence of impulsive noise. Beyond this, the EAPA algorithm demonstrates robust adaptive capabilities across diverse impulsive noise conditions without requiring prior information, further validating its generalization performance.

b) Simulation Experiment 2: Performance comparison of EAPA versus SCM [9], ZMNL [11], and INCK [14] under different signal-to-noise ratios and impulsive noise conditions.

For the simulation verification, we selected four algorithms—SCM, ZMNL, INCK, and EAPA—for comparative analysis. Two far-field narrowband independent signals with

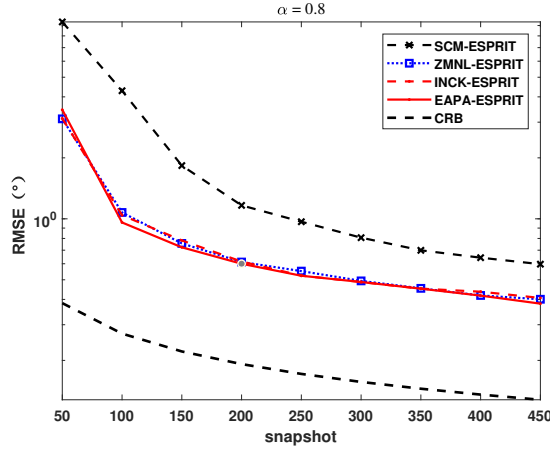


Fig. 4. Performance comparison of four algorithms at different snapshot rates When $\alpha=0.8$.

azimuth angles $[-5^\circ, 0^\circ]$ were incident onto a 10-element uniform linear array, with a fast-snap count $T = 600$. Impulsive noise levels were set to $\alpha = 0.8$ and $\alpha = 1.5$, with GSNR varying from 0 dB to 8 dB. Through 1000 MC, the DOA estimation performance of each algorithm was evaluated under different generalized GSNR conditions.

As observed in Fig. 2, when $\alpha = 0.8$, the EAPA algorithm demonstrates superior performance compared to the SCM, ZMNL, and INCK algorithms. This is because when the GSNR exceeds 3, the EAPA method and INCK become comparable. Although the INCK algorithm appears lower on the curve, it possesses precise prior noise information, granting it an advantage in high impulsive noise scenarios, whereas the EAPA algorithm requires no prior information. When the GSNR exceeds 3, the EAPA algorithm remains at the lowest point without requiring prior information, further demonstrating its superiority.

As illustrated in Fig. 3, when $\alpha = 1.5$, RMSE of the EAPA algorithm is significantly lower than that of the SCM and ZMNL algorithms under identical GSNR conditions. Particularly in the low GSNR range, the EAPA algorithm demonstrated superior noise suppression capability and angle estimation stability, further validating its enhanced robustness and higher estimation accuracy under non-Gaussian impulsive noise conditions. Although the RMSE curves of INCK and EAPA were very close when the signal-to-noise ratio exceeded 6 dB, the INCK algorithm relies on precise prior noise information. In contrast, the EAPA algorithm achieves adaptive suppression of impulsive noise based solely on the received signal, without requiring any prior knowledge of noise statistics. Therefore, EAPA holds a distinct overall advantage.

c) Simulation Experiment 3: Performance comparison of the EAPA algorithm with other algorithms under different impulsive noise and fast-sorting numbers conditions.

In simulation verification, four algorithms were selected for comparative analysis. Two far-field narrowband independent

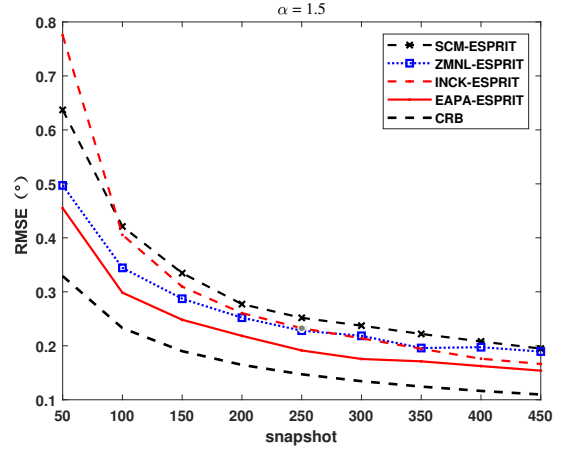


Fig. 5. Performance comparison of four algorithms at different snapshot rates When $\alpha=1.5$.

signals were incident on a 10-element uniform line array at azimuth angles $[-5^\circ, 0^\circ]$. The generalized signal-to-noise GSNR was set to 5 dB, with impulsive noise levels of $\alpha = 0.8$ and $\alpha = 1.5$. The number of snapshots varied from $T = 50$ to $T = 450$. Through MC = 1000 monte carlo simulations, the DOA estimation performance of each algorithm under different snapshot conditions was evaluated.

As shown in the simulation results in Fig. 4, when $\alpha = 0.8$, the estimation performance of the EAPA algorithm is comparable to that of the ZMNL and INCK algorithms. However, it is noteworthy that without relying on any prior noise statistics, the EAPA algorithm still achieves the lowest RMSE in most experimental scenarios.

Under the condition of $\alpha = 1.5$, the EAPA algorithm demonstrates significantly greater advantages than traditional methods in suppressing impulsive noise and enhancing estimation accuracy. As shown in Fig. 5, under identical snapshot conditions, the RMSE curve of the EAPA algorithm consistently lies below those of the SCM and INCK comparison algorithms. This comparative result further confirms that the EAPA algorithm exhibits superior noise robustness in complex environments with non-Gaussian noise, while also significantly enhancing the accuracy and stability of direction-of-arrival estimation.

V. CONCLUSION

To address the limitation of traditional algorithms requiring prior noise information, this paper proposes the EAPA algorithm model. It introduces information entropy and reconstructs the kernel function, controlling the kernel's effectiveness through the magnitude of information entropy. This achieves adaptive handling of impulsive noise, enabling DOA estimation under impulsive noise conditions without requiring prior noise information. Simulation results demonstrate that the EAPA algorithm overcomes the limitation of requiring prior noise information, exhibiting significant advantages across different snapshot and GSNR.

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